

Resilient Secure Control of Networked Industrial Control Systems Under AI-Driven Feedback Poisoning via ACK Bundling

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Abstract—This paper investigates resilient secure control for networked industrial systems under AI-driven feedback poisoning attacks. Such attacks exploit feedback channels and timing regularities to degrade closed-loop performance while remaining difficult to detect. A data-driven poisoning model is constructed based on a real-world SWaT dataset, enabling the emulation of adversarial feedback manipulation beyond idealized assumptions. An acknowledgment (ACK) bundling mechanism is introduced to regulate feedback transmission by aggregating ACKs within a tunable time window. The bundling window T_b is adaptively adjusted via gradient descent (GD), facilitating a flexible trade-off between control performance and communication efficiency in adversarial environments. In addition, a lightweight Lyapunov-Krasovskii functional (LKF) is developed to establish closed-loop stability in the presence of communication delays and poisoned feedback. Simulation results on a networked unmanned surface vehicle (USV) platform demonstrate improved tracking performance and robustness against feedback poisoning under constrained communication

conditions. These results indicate that the proposed framework provides a practical and effective solution for secure networked industrial control.

Index Terms—Industrial control systems 4.0, Lyapunov-Krasovskii functional, machine learning, ACK bundling, cyber-attacks.

I. INTRODUCTION

THE 21st century has witnessed a global surge in industrial modernization, driven by initiatives in the United States, Germany, and China to reshape traditional manufacturing systems [1], [2]. A central enabler of this transformation is the *Industrial Internet*, which redefines how industrial control systems (ICSs) interact with modern information technologies. Supported by advances in artificial intelligence (AI), fifth-generation (5G) communication, edge computing, cloud services, and Big Data analytics [3], industrial networks have become the operational backbone of ICSs 4.0. Such networks are expected to support massive device connectivity, heterogeneous interoperability, and deterministic low-latency communication [4], placing ICSs 4.0 in increasingly interconnected networked control environments.

Meanwhile, control architectures have evolved from centralized to distributed and fieldbus-based structures with advances in computation, communication, and control technologies [5]. As industrial and information technologies further converge, ICSs are increasingly connected to public and open networks, forming ICSs 4.0. These systems integrate sensing, computation, and actuation to automate industrial processes, monitor key variables, and enable real-time decision-making [6], [7]. This increasing openness also expands the attack surface. National strategies, such as Germany's *Industrie 4.0*, the United States' *Industrial Internet*, and China's *Made in China 2025*, have accelerated the interconnection of ICSs with enterprise information technology systems, cloud platforms, and third-party services [8], thereby increasing exposure to vulnerabilities in Internet-of-Things (IoT) infrastructures, cloud environments, and large-scale data-sharing.

Recent cyber incidents, such as the 2018 *WannaCry* attack on TSMC [9] and the 2023 Colonial Pipeline breach [10], [11], have highlighted the growing security risks faced by modern industrial infrastructures [12]. Beyond conventional ransomware threats, AI-driven data poisoning attacks pose a more subtle

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and critical challenge to ICSs 4.0. By manipulating sensor measurements and exploiting control-loop dynamics, such attacks can gradually degrade control performance while remaining difficult to detect using traditional anomaly-detection mechanisms. The leakage of production parameters and control logic further exacerbates these risks, threatening operational continuity and system safety [13]. As a result, enhancing resilience against intelligent and data-driven cyber threats has become a pressing research problem in networked industrial control systems.

II. RELATED WORK

In this section, existing works on resilient control and data poisoning defense are briefly reviewed to contextualize the contributions of this paper. In the domain of ICS security, resilient and secure control strategies have been extensively studied under various cyber disturbances and network uncertainties. Existing works mainly focus on preserving stability and control performance through robust control design and proactive compensation mechanisms. For instance, observability-preserving control frameworks [14], learning-assisted monitoring strategies under denial-of-service (DoS) attacks [15], [16], [17], and fuzzy control schemes with elastic event-triggered mechanisms [18], [19] have been proposed to mitigate network-induced disruptions. While these methods are effective against random disturbances and predefined attack patterns, they generally cannot explicitly capture intelligent data poisoning behaviors that exploit feedback manipulation and timing predictability in closed-loop systems.

In parallel, learning-based detection and defense techniques have been developed to enhance attack awareness in ICSs. Representative approaches include game-theoretic attack prediction [20], blockchain-enabled trustworthy data sharing [21], and deep learning-based intrusion detection methods [22], [23]. Despite their effectiveness in identifying abnormal behaviors, these mechanisms are typically deployed as external or supervisory layers and are not tightly integrated with the control loop [24]. Consequently, their protection capability may be limited against stealthy data poisoning attacks that directly corrupt feedback signals in real time. Moreover, existing communication-efficient control and resource allocation strategies [25] primarily target bandwidth reduction or energy efficiency under benign network conditions, and seldom address adversarial timing manipulation within secure control design.

Despite the existing progress, important gaps remain in both theory and practice. Most current resilient control studies still rely on simplified attack assumptions and do not explicitly address AI-driven feedback poisoning with realistic temporal characteristics [26]. Meanwhile, many existing defense schemes focus on either attack detection or communication efficiency alone, without jointly considering adaptive feedback regulation and closed-loop stability under adversarial conditions [27]. This gap is especially relevant to ICSs 4.0, where the openness of networked feedback loops increases both attack exposure and communication-control coupling [28]. Therefore, a unified framework integrating realistic poisoning modeling, adaptive

TABLE I
NOTATIONS AND DESCRIPTION

Symbol	Description
$\mathbb{R}^n$	The n -dimensional Euclidean space.
$\mathbb{R}^{m \times n}$	The set of all $m \times n$ real matrices.
$Q > 0$	A symmetric positive definite matrix Q .
A^{-1}	The inverse of matrix A .
A^T	The transpose of matrix A .
I	Identity matrix of appropriate dimensions.
$L_2[0, +\infty)$	The space of square-summable infinite vector sequences.
$\text{He}\{Q\}$	$\text{He}\{Q\} = Q + Q^T$.
$\text{diag}\{\cdots\}_n$	Block diagonal matrix with n blocks.

communication regulation, and stability-guaranteed resilient control is still needed.

Intelligent data poisoning and timing-aware feedback manipulation significantly threaten the reliability of networked ICSs 4.0. The distinguishing aspect of the proposed approach lies in the unified integration of SWaT-trace-driven attack activation modeling, adaptive ACK bundling, GD-based online adjustment, and LKF-based stability analysis. Unlike existing studies that typically address communication regulation, attack mitigation, or controller adaptation separately, the proposed design jointly considers communication efficiency, online adaptability, and closed-loop stability under adversarial network conditions.

The main contributions of this paper are summarized as follows:

- A data-driven poisoning model is constructed using the SWaT dataset, enabling more realistic evaluation of feedback manipulation in ICSs 4.0 than idealized attack assumptions.
- An adaptive ACK bundling mechanism is developed to regulate feedback transmission in a communication-aware manner, thereby reducing communication overhead while mitigating timing-based vulnerabilities.
- A unified resilient secure control framework is established by coupling GD-based online adjustment with a simplified LKF, so as to jointly address online adaptability, communication efficiency, and closed-loop stability under adversarial uncertainty.
- Extensive simulations on a networked USV platform demonstrate the effectiveness of the proposed method in achieving improved tracking accuracy, enhanced delay tolerance, and strong robustness against intelligent data poisoning attacks.

III. PRELIMINARIES

Before presenting the system model, the mathematical notations used throughout this paper are summarized in Table I. For clarity, the proposed secure framework under feedback poisoning attacks is illustrated in Fig. 1

A. ICSs 4.0 and Dislocation Communication Controller

The considered ICSs 4.0 are modeled by the following continuous-time state-space dynamics:

$$\dot{x}(t) = Ax(t) + Bu(t) + C\omega(t), \quad (1)$$

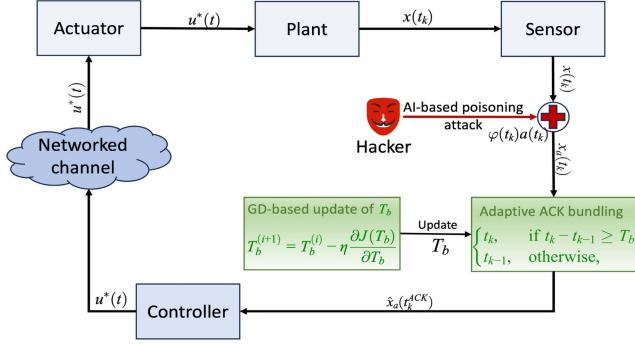


Fig. 1. Secure framework under feedback poisoning attacks.

where $x(t)$ denotes the system state vector, $u(t)$ is the control input, and $\omega(t)$ represents external disturbances. The constant matrices A , B , and C describe the nominal system dynamics.

In ICSs 4.0, communication delays, packet losses, and asynchronous updates commonly arise due to heterogeneous sensing, actuation, and networking infrastructures. Such imperfections can significantly degrade the performance of conventional synchronous control schemes. A dislocation communication controller is adopted, which explicitly incorporates delayed state feedback to compensate for temporal misalignments in the control loop and enhance closed-loop robustness. Control updates are generated via a zero-order hold (ZOH) mechanism, yielding the sampling sequence

$$\mathbb{S} = \{t_0, t_1, \dots, t_k, \dots\}, \quad k = 0, 1, 2, \dots,$$

where the sampling interval $h = t_k - t_{k-1}$ satisfies $0 \leq h_m \leq h \leq h_M$. The dislocation-based control input is formulated as

$$u(t) = Kx(t_k) + Kx(t_k - d), \quad t \in [t_{k-1}, t_k), \quad (2)$$

where K denotes the state-feedback gain matrix to be designed, and $d \leq h_M$ represents a bounded communication delay. By simultaneously exploiting current and delayed state information, this control structure enhances robustness against network-induced uncertainties and temporal disturbances.

B. AI-Driven Data Poisoning Attack Model

The attacker is assumed to operate within a networked industrial control environment and can inject corrupted sensor measurements and manipulate feedback timing information over the communication channel. Rather than triggering abrupt system failures, the attack aims to stealthily degrade closed-loop control performance by exploiting feedback dependencies and timing predictability. The adversary is subject to practical industrial constraints, including limited access to sensing and actuation channels, restricted intervention frequency, and temporal patterns consistent with real-world cyber-physical attacks. Under these conditions, the attack primarily manifests as data poisoning and timing-aware feedback corruption that affects both real-time and delayed control information. The key elements of the considered threat model are summarized in Table II.

TABLE II
THREAT MODEL FOR AI-DRIVEN DATA POISONING ATTACKS

Aspect	Description
Capability	Partial injection of corrupted sensor data and manipulation of feedback timing.
Objective	Gradual degradation of control performance with stealthiness.
Attack surface	Sensor measurements and networked feedback channels.
Attack strategy	Data poisoning and timing-aware feedback corruption.
Constraints	Partial channel access and realistic temporal patterns derived from SWaT.

TABLE III
EXCERPT OF SWaT ATTACK SCENARIOS

ID	Attack point	Duration (min)	Description
1	MV-101	15.7	Close valve $\rightarrow$ overflow
2	P-102	7.4	Force pump $\rightarrow$ burst risk
3	LIT-101	6.4	Raise level $\rightarrow$ mislead
4	MV-504	6.5	Open valve $\rightarrow$ RO failure
5	—	—	No observable effect
6	AIT-202	3.3	Fix pH $\rightarrow$ downstream error
7	LIT-301	7.1	Raise level $\rightarrow$ pump damage
8	DPIT-301	16.0	Raise pressure $\rightarrow$ loop fault
9	—	—	No observable effect
10	FIT-401	2.7	Lower flow $\rightarrow$ UV off

AI-driven data poisoning attacks pose a growing threat to ICSs4.0 operating over networked feedback loops, as they exploit system dynamics and timing information to induce stealthy performance degradation. In this work, such adversarial behaviors are characterized using real-world cyber-physical attack traces from the SWaT dataset [29]. Compared with purely artificial attack assumptions or randomly generated activation patterns, the use of SWaT-derived traces enables the poisoning model to better reflect realistic attack occurrence patterns in industrial environments. Meanwhile, to incorporate these data-driven attack patterns into the control framework in a compact yet analyzable manner, a binary activation indicator $\varphi(t_k)$ is introduced to denote whether adversarial interference is present at the sampling instant t_k . This treatment preserves the essential temporal information of attack activation while avoiding excessive modeling complexity, thereby facilitating the subsequent resilient control design and stability analysis:

$$\varphi(t_k) = \begin{cases} 1, & \text{attack active,} \\ 0, & \text{attack dormant.} \end{cases} \quad (3)$$

The SWaT dataset contains 41 labeled attack scenarios collected from a real-world water treatment system, with annotations on timing, targets, manipulation types, and physical outcomes. The attacks span sensor spoofing (e.g., LIT, FIT, AIT), actuator overrides, logic interference, and stealthy coordination. Table III summarizes representative cases, highlighting diverse manipulation strategies and their dynamic impact on the underlying physical process.

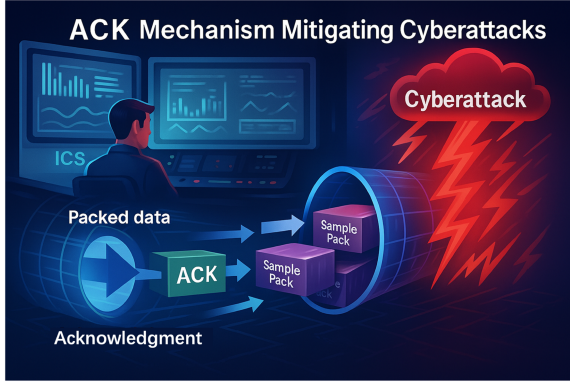


Fig. 2. ACK bundling mechanism under feedback attacks.

Most scenarios in the SWaT dataset target critical sensing or actuation components, leading to cascading physical effects such as overflows, pressure surges, or process deadlocks. These attacks exhibit strong coupling between adversarial manipulation and system-level degradation, making SWaT a suitable benchmark for evaluating secure control under realistic cyber-physical disruptions. To support control-oriented simulation, the raw attack traces are aligned with the system's sampling period and normalized against nominal operating ranges. A binary switching indicator $\varphi(t_k)$ denotes attack activation at each sampling instant, where $\varphi(t_k) = 1$ indicates that the attack is active. The processed signals are then injected into the closed-loop dynamics, enabling a systematic assessment of controller performance under timing-aware data poisoning.

Based on the activation signal $\varphi(t_k)$, the corrupted control input under attack is modeled as

$$u^*(t) = \varphi(t_k)Kx(t_k) + \varphi(t_k)Kx(t_k - d). \quad (4)$$

This formulation captures the simultaneous corruption of both real-time and delayed feedback information in the closed-loop system.

Remark 1: The poisoning behaviors in this work are not artificially designed but derived directly from temporally correlated attack traces in the SWaT dataset, enabling reproducible, data-driven evaluation under realistic conditions. While the USV platform does not replicate the physical processes of the SWaT water plant, it serves as a representative networked control system with sensing, actuation, and feedback communication. The adversarial patterns from SWaT are introduced at the communication layer, decoupled from plant-specific dynamics, to evaluate feedback resilience. This design ensures that the proposed ACK bundling and gradient-based adaptation strategies remain broadly applicable to various ICS 4.0 architectures with feedback-driven loops, provided that appropriate system-level parameters are adopted.

C. Acknowledgment Bundling Mechanism

Figs. 2 and 3 illustrate the contrast between conventional ACK exchange and the proposed bundling mechanism. In standard ICS communication, frequent ACKs confirm the delivery of

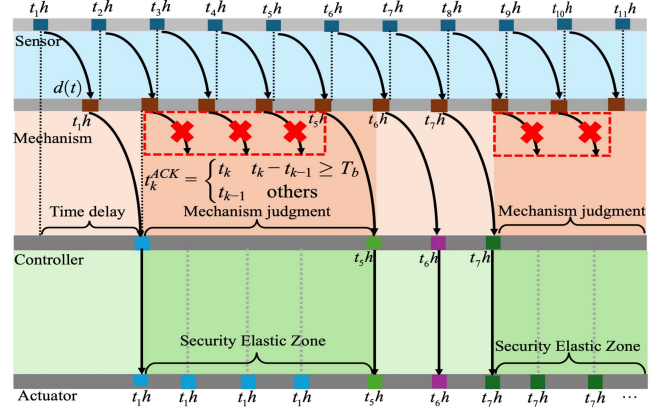


Fig. 3. Operating principle of the ACK bundling mechanism.

sensing and control packets, but this regularity introduces overhead and timing predictability that adversaries can exploit [30]. To mitigate such vulnerabilities, this work adopts an ACK bundling strategy: ACKs are transmitted once per bundling window T_b , aggregating multiple events into a single response. This reduces feedback frequency and disrupts temporal patterns at the communication layer. The resulting partial observability and delay are explicitly incorporated into the closed-loop model and stability analysis, ensuring that control performance is preserved under reduced update rates. The ACK transmission rule is defined as follows:

$$t_k^{\text{ACK}} = \begin{cases} t_k, & \text{if } t_k - t_{k-1} \geq T_b, \\ t_{k-1}, & \text{otherwise,} \end{cases} \quad (5)$$

where t_k^{ACK} denotes the ACK transmission time, t_k is the current sampling instant, and T_b represents the bundling window. An ACK is issued only when the elapsed interval since the previous transmission exceeds T_b ; otherwise, the most recent ACK is reused. By sparsifying ACK updates, the proposed mechanism reduces network load while injecting controlled temporal uncertainty into the feedback loop, thereby hindering adversarial synchronization. Owing to its lightweight structure, the ACK bundling mechanism can be readily integrated into existing industrial communication infrastructures.

Remark 2: ACK bundling, as used in this work, serves as a communication-layer mechanism to modulate feedback frequency under adversarial conditions. The core contribution lies not in the bundling itself, but in its dynamic adaptation via a gradient-descent strategy that responds to real-time feedback perturbations. This design enables online adjustment of the bundling window T_b to balance control responsiveness and communication load, introducing timing variability that disrupts adversarial inference. Unlike fixed-rate or manually tuned schemes, the proposed method tightly couples adaptive communication with control stability analysis, ensuring resilient performance under delay and feedback conditions.

D. Gradient Descent Optimization of Bundling Window

Adapting the ACK bundling mechanism to dynamic environments requires treating the window T_b as a tunable variable rather than a fixed parameter. A GD algorithm is employed to iteratively adjust T_b based on real-time system feedback [31], so as to balance control performance and communication efficiency under adversarial interference. This approach enables continuous refinement of the ACK policy in delay-sensitive and uncertain settings. A composite cost function is introduced to guide the adjustment of T_b :

$$J(T_b) = \alpha \sum_{k=1}^N \|x_k(T_b) - x_{\text{ref}}\|^2 + \beta \cdot \frac{N}{T_b}, \quad (6)$$

where $\alpha, \beta > 0$ are weighting factors. The first term penalizes control degradation due to delayed feedback, while the second term penalizes communication frequency.

The two components of the cost function are defined as follows:

$$J_{\text{ctrl}}(T_b) = \sum_{k=1}^N \|x_k(T_b) - x_{\text{ref}}\|^2, \quad (7)$$

$$J_{\text{comm}}(T_b) = \frac{N}{T_b}, \quad (8)$$

where $x_k(T_b)$ represents the system state at the k -th instant under bundling window T_b , and x_{ref} denotes the desired reference state.

Since the system state x_k implicitly depends on the bundling window T_b through the closed-loop dynamics, obtaining an analytical expression for the gradient is generally intractable. To enable practical implementation, the gradient is approximated using a forward finite-difference scheme [32]:

$$\frac{\partial x_k}{\partial T_b} \approx \frac{x_k(T_b + \delta) - x_k(T_b)}{\delta}, \quad (9)$$

where δ is a small perturbation constant. Substituting this approximation into the cost function yields the following expression:

$$g_{\text{ctrl}}(T_b) = \sum_{k=1}^N \frac{2(x_k(T_b) - x_{\text{ref}})^T (x_k(T_b + \delta) - x_k(T_b))}{\delta},$$

$$\frac{\partial J(T_b)}{\partial T_b} \approx \alpha \cdot g_{\text{ctrl}}(T_b) - \beta \cdot \frac{N}{T_b^2}. \quad (10)$$

The bundling window T_b is then updated according to the standard GD rule:

$$T_b^{(i+1)} = T_b^{(i)} - \eta \cdot \frac{\partial J(T_b)}{\partial T_b}, \quad (11)$$

where $\eta > 0$ denotes the learning rate and i is the iteration index. The overall adaptive procedure is summarized in Algorithm 1. Notably, this update scheme allows T_b to evolve in response to observed system behavior without requiring closed-form gradient expressions, which is particularly well-suited for networked control settings with uncertain dynamics and adversarial interference.

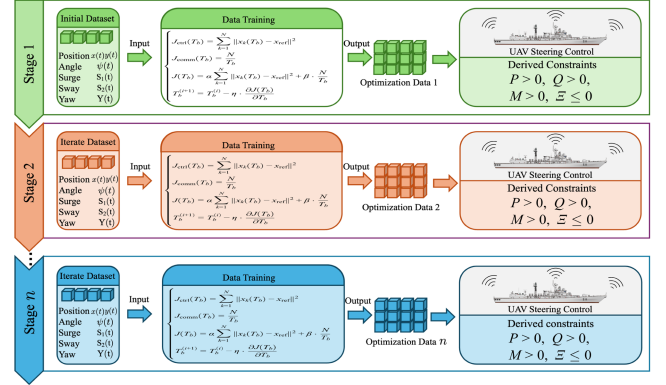


Fig. 4. Adaptive optimization of the bundling window T_b .

Remark 3: In the presence of complex closed-loop dynamics and uncertain delays, the cost function exhibits implicit and nonlinear dependence on the bundling window T_b . As analytical gradients are intractable, this work adopts a forward finite-difference approximation based on observed cost changes. Compared with model-based gradient evaluation or more elaborate approximation schemes, this treatment avoids explicit model differentiation and reduces online computational burden while still preserving the sensitivity information needed for adaptive window regulation. Therefore, it offers a practical balance among implementation simplicity, online adaptability, and analytical tractability under adversarial real-time network conditions.

Remark 4: As shown in Fig. 4, unlike static ACK strategies with fixed or heuristic settings, the proposed method adjusts T_b through closed-loop gradient-based optimization. By embedding both control accuracy and communication load into the cost function, the bundling frequency self-adapts to current system conditions. This coupling enhances resilience against adversarial timing interference and improves adaptability to time-varying network states. Meanwhile, inappropriate parameter tuning may lead to excessively slow adjustment or oscillatory T_b evolution, which can degrade transient performance and reduce the closed-loop stability margin. To mitigate such effects, the proposed design combines moderated online adaptation with LKF-based stability analysis, so that communication regulation and control performance can be balanced within a feasible operating range.

Then, the ICSs 4.0 (1) can be transformed into the following:

$$\dot{x}(t) = Ax(t) + \varphi(t_k)B[Kx(t_k) + Kx(t_k - d)] + C\omega(t), \quad (12)$$

where $t \in [t_{k-1}, t_k^{ACK})$ and t_k^{ACK} is supervised by the ACK window mechanism and optimized by the GD algorithm.

Lemma 1: (Jensen's inequality [33]) For matrix $S > 0$, there exists a function $x(s)$ such that the following integral inequality is satisfied over the interval $[n, m]$ in the space $\mathbb{R}^{n \times n}$:

$$(m - n) \int_n^m x^\top(s) S x(s) ds$$

Algorithm 1: Gradient-Based Optimization of Window T_b .

Require: Initial window $T_b^{(0)} > 0$, learning rate η ,
perturbation δ , horizon N , weights α, β

Ensure: Optimized window T_b^*

- 1: Initialize $T_b \leftarrow T_b^{(0)}$
- 2: **for** $i = 1$ to M **do**
- 3: Simulate system with T_b to obtain state trajectory
 $\{x_k(T_b)\}_{k=1}^N$
- 4: Compute performance loss:

$$J_{\text{ctrl}}(T_b) \leftarrow \sum_{k=1}^N \|x_k(T_b) - x_{\text{ref}}\|^2$$

- 5: Compute communication loss:

$$J_{\text{comm}}(T_b) \leftarrow \frac{N}{T_b}$$

- 6: $J(T_b) \leftarrow \alpha \cdot J_{\text{ctrl}}(T_b) + \beta \cdot J_{\text{comm}}(T_b)$
- 7: Simulate system with $T_b + \delta$ to get $\{x_k(T_b + \delta)\}_{k=1}^N$
- 8: Approximate gradient of $J_{\text{ctrl}}(T_b)$:

$$g_{\text{ctrl}}(T_b) \leftarrow \sum_{k=1}^N \frac{2(x_k(T_b) - x_{\text{ref}})^T (x_k(T_b + \delta) - x_k(T_b))}{\delta}$$

- 9: Approximate full gradient:

$$\frac{\partial J(T_b)}{\partial T_b} \leftarrow \alpha \cdot g_{\text{ctrl}}(T_b) - \beta \cdot \frac{N}{T_b^2}$$

- 10: Update window:

$$T_b \leftarrow T_b - \eta \cdot \frac{\partial J(T_b)}{\partial T_b}$$

- 11: **if** $|T_b^{(i)} - T_b^{(i-1)}| < \varepsilon$ **then**
- 12: **break**
- 13: **end if**
- 14: **end for**
- 15: **return** $T_b^* \leftarrow T_b$

$$\geq \left(\int_n^m x(s) ds \right)^\top S \left(\int_n^m x(s) ds \right).$$

IV. PRELIMINARIES

Theorem 1: Consider the ICSs 4.0 system described by (12). Suppose that the system experiences a bounded communication delay $d > 0$. If the system is asymptotically stable, then there exist symmetric positive definite matrices P, Q , and M , along with arbitrary matrices N_1 and N_2 , such that the following LMI conditions hold:

$$P > 0, \quad Q > 0, \quad M > 0, \quad \Xi \leq 0, \quad (13)$$

Where

$$\xi(t) = \{x(t), x(t_k), x(t_k^{\text{ACK}}), x(t-d), \dot{x}(t), \omega(t)\},$$

$$\Xi = \Sigma + \text{He}\{\Lambda\Gamma\}, \quad \Sigma = \text{He}\{e_1^\top P e_5\} + e_3^\top Q e_3$$

$$- e_1^\top Q e_1 + d^2 e_5^\top M e_5 - \Phi^\top M \Phi,$$

$$\Lambda = e_2^\top N_1 + e_6^\top N_2, \quad \Phi = \text{col}\{e_1 - e_4\},$$

$$\Gamma = A e_1 + B K e_1 + B K e_4 + C e_6 - e_5,$$

$$e_i = [0_{n \times (i-1)n} \quad I_{n \times n} \quad 0_{n \times (6-i)n}], \quad i = 1, 2, \dots, 6.$$

Proof: To prove the stability of the closed-loop system, we construct a set of LKFs of the form:

$$V(t) = \sum_{i=1}^3 V_i(t), \quad t \in [t_{k-1}, t_k^{\text{ACK}}), \quad (14)$$

where the individual terms are defined as follows:

$$V_1(t) = x^\top(t) P x(t),$$

$$V_2(t) = \int_t^{t_k^{\text{ACK}}} x^\top(s) Q x(s) ds,$$

$$V_3(t) = d \int_{t-d}^t \int_\theta^t \dot{x}^\top(s) M \dot{x}(s) ds d\theta.$$

According to Lyapunov stability theory, we evaluate the time derivative of the total functional:

$$\frac{dV(t)}{dt} = \sum_{i=1}^3 \frac{dV_i(t)}{dt}.$$

Each component is derived as:

$$\frac{dV_1(t)}{dt} = \text{He} \{x^\top(t) P \dot{x}(t)\}, \quad (15)$$

$$\frac{dV_2(t)}{dt} = x^\top(t_k^{\text{ACK}}) Q x(t_k^{\text{ACK}}) - x^\top(t) Q x(t), \quad (16)$$

$$\frac{dV_3(t)}{dt} = d^2 \dot{x}^\top(t) M \dot{x}(t) - d \int_{t-d}^t \dot{x}^\top(s) M \dot{x}(s) ds. \quad (17)$$

Applying Jensen's inequality from Lemma 1 to the integral term in (17) yields the following upper bound:

$$\begin{aligned} & -d \int_{t-d}^t \dot{x}^\top(s) M \dot{x}(s) ds \\ & \leq - \left(\int_{t-d}^t \dot{x}(s) ds \right)^\top M \left(\int_{t-d}^t \dot{x}(s) ds \right). \end{aligned} \quad (18)$$

Combining (15)–(18), the derivative of the constructed LKF can be written as

$$\sum_{i=1}^3 \frac{dV_i(t)}{dt} = \xi^\top(t) \Sigma \xi(t). \quad (19)$$

Next, for the closed-loop system in (12), two auxiliary matrices N_1 and N_2 are introduced to incorporate the closed-loop dynamics into the same quadratic form:

$$\begin{aligned} 0 &= 2 [x^\top(t_k) N_1 + \omega^\top(t) N_2] [Ax(t) + \varphi(t_k) \\ & \quad \times B(Kx(t) - Kx(t-d)) + C\omega(t) - \dot{x}(t)] \\ &\leq 2 [x^\top(t_k) N_1 + \omega^\top(t) N_2] [Ax(t) + BKx(t) \\ & \quad - BKx(t-d) + C\omega(t) - \dot{x}(t)] \end{aligned}$$

$$= \xi^\top(t) \text{He}\{\Lambda\Gamma\} \xi(t). \quad (20)$$

Therefore, the total LKF derivative becomes:

$$\sum_{i=1}^3 \frac{dV_i(t)}{dt} = \xi^\top(t) \Xi \xi(t), \quad (21)$$

where $\Xi = \Sigma + \text{He}\{\Lambda\Gamma\}$.

Under the condition defined in (13), it holds that:

$$\sum_{i=1}^3 \frac{dV_i(t)}{dt} = \xi^\top(t) \Xi \xi(t) < 0, \quad (22)$$

which guarantees the asymptotic stability of the ICSs 4.0 described in (12).

Remark 5: The proposed framework is intentionally designed as an integrated structure rather than a collection of independent components. In this setting, realistic attack modeling determines how feedback corruption enters the control loop, ACK bundling shapes the communication behavior exposed to such attacks, and gradient-based tuning governs how this interaction evolves online. The LKF analysis then provides a unified stability guarantee for the resulting closed-loop dynamics. This coupling ensures that security, communication efficiency, and stability are addressed within a single control loop rather than as loosely connected design objectives.

Theorem 2: Consider the ICSs 4.0 system described by (12). Suppose the system admits a bounded delay $d > 0$ and a positive design parameter $\mu > 0$. If the system is asymptotically stable, then there exist symmetric positive definite matrices $\hat{P}$, $\hat{Q}$, and $\hat{M}$, and auxiliary matrices X and Y , such that the following LMI conditions hold:

$$\hat{P} > 0, \quad \hat{Q} > 0, \quad \hat{M} > 0, \quad \hat{\Xi} \leq 0, \quad (23)$$

Where

$$\begin{aligned} \hat{\Xi} &= \hat{\Sigma} + \text{He}\{\hat{\Lambda}\hat{\Gamma}\}, \quad \hat{\Lambda} = e^\top(t)_2 + \mu e^\top(t)_6, \\ \hat{\Sigma} &= \text{He}\{e_1^\top \hat{P} e_5\} + e_3^\top \hat{Q} e_3 - e_1^\top \hat{Q} e_1 \\ &\quad + d^2 e_5^\top \hat{M} e_5 + \Phi^\top(t) \hat{M} \Phi, \end{aligned}$$

Proof: Further gain matrices are formulated as follows:

$$\begin{aligned} \hat{\Gamma} &= AXe_1 + BYe_1 + BYe_4 + CXe_6 - Xe_5, \\ K &= YX^{-1}, N_1 = X^{-1}, N_2 = \mu X^{-1}, \\ \hat{P} &= X^\top PX, \quad \hat{Q} = X^\top QX, \quad \hat{M} = X^\top MX. \end{aligned}$$

Premultiplying and postmultiplying the inequality in (13) by $X^\top$ and X , respectively, transforms the condition into the form given in (23). The remaining derivation follows the same LKF-based argument as in Theorem 1 and is omitted for brevity. ■

V. ILLUSTRATIVE EXAMPLE

The proposed framework is validated on a Simulink-based simulation platform emulating a networked USV, as illustrated in Fig. 5. The USV integrates onboard sensing, actuation, and real-time feedback over a communication network. Control inputs

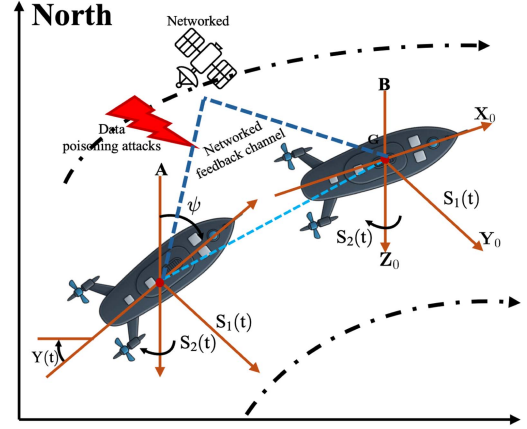


Fig. 5. USV trajectory tracking under feedback poisoning.

are generated by a remote controller employing ACK bundling and gradient-based adaptation, thereby forming a closed-loop system. To assess resilience under intelligent adversarial interference, real-world attack traces from the SWaT dataset [29] are injected into the feedback channel. Although originally collected from water treatment systems, these traces capture realistic poisoning patterns, such as sensor spoofing and timing-aware feedback corruption, and are applicable to general networked control settings. The USV testbed serves as a representative scenario for evaluating feedback-layer defenses rather than for modeling a specific industrial process. The proposed ACK bundling mechanism is particularly suitable for networked ICSs with communication-coupled feedback loops, such as industrial process control, supervisory control, and distributed automation systems, where communication efficiency and control performance are closely coupled. Plant dynamics and control mappings follow benchmark configurations in [34].

A structured evaluation is conducted through three experimental modules to validate the proposed framework from complementary perspectives:

- **Part A:** Investigation of the influence of the learning rate η on the convergence behavior of the bundling window T_b and the resulting closed-loop dynamics, with emphasis on parameter sensitivity and online adaptability.
- **Part B:** Evaluation of control performance under varying communication delays, focusing on delay tolerance, transient behavior, and the robustness of the communication-aware regulation mechanism.
- **Part C:** Assessment of closed-loop resilience under AI-driven feedback poisoning attacks, including the effects of corrupted feedback on trajectory tracking, ACK adaptation, and control smoothness.

Together, these experiments provide a structured validation of the proposed method under parameter variation, network impairment, and adversarial feedback conditions. Unless otherwise specified, all experiments employ a fixed state-feedback controller designed via LMIs. The sampling period is selected from $h \in \{0.01, 0.05, 0.1\}$, and the communication delay varies up to $d = 2.319$. The ACK bundling window is initialized within

TABLE IV
TIME DELAY AND T_b UNDER THE GD ALGORITHM

Learning Rate η	Max Delay d	T_b Convergence Behavior
0.001	2.417	Slow and smooth
0.005	2.319	Moderate and stable
0.100	2.269	Fast, prone to fluctuation
0.500	2.103	Aggressive, reduced robustness

TABLE V
COMMUNICATION SIGNAL NUMBERS UNDER DIFFERENT SAMPLING PERIODS h

h	Refs.		This Paper (Normal)		This Paper (Attack)	
	[34]	[35]	Fixed	Gradient	Fixed	Gradient
0.01	50	67	42	38	47	43
0.05	39	52	39	32	41	38
0.10	30	47	33	22	37	30

a feasible range and updated online using a GD scheme with learning rates $\eta \in \{0.001, 0.005, 0.1, 0.5\}$. Adversarial feedback corruption is injected using SWaT-based attack traces aligned with the system sampling instants.

A. System Supervised by the GD Algorithm

This part evaluates the robustness of the proposed ACK bundling mechanism under gradient-based supervision, with emphasis on stability margins and communication tolerance rather than learning performance. A fixed state-feedback controller is used throughout all experiments to ensure consistency across different test conditions. The feedback gain matrix is computed offline by solving LMIs using the YALMIP toolbox. Under nominal operating conditions with a communication delay $d = 0.3$ and weighting factor $\mu = 0.5$, the resulting controller gain is $K = [6.0721, -2.4114, -16.6314, 9.3121, -26.4650, -0.2826]$.

To examine the impact of adaptive ACK regulation on communication tolerance, different learning rates η are selected as implementation parameters of the gradient update. For each setting, the maximum admissible communication delay that preserves closed-loop stability is identified. This delay margin characterizes the system's robustness to timing uncertainty under ACK bundling and is influenced by the smoothness of the T_b adaptation process. Representative results are summarized in Table IV. The results indicate that overly small step sizes slow down the transient adjustment of T_b , while overly large ones introduce noticeable oscillations. A moderate η yields a smoother transient and a competitive delay margin and is used in the following experiments unless otherwise stated.

Table V compares the number of communication signals generated under different sampling periods h for representative communication-efficient baselines and the proposed ACK-based schemes under both nominal operation and data poisoning attacks. The reference methods [34], [35] adopt event-triggered or communication-reduction mechanisms under fixed control architectures and thus serve as appropriate baselines for evaluating communication efficiency at the feedback layer. Learning-based

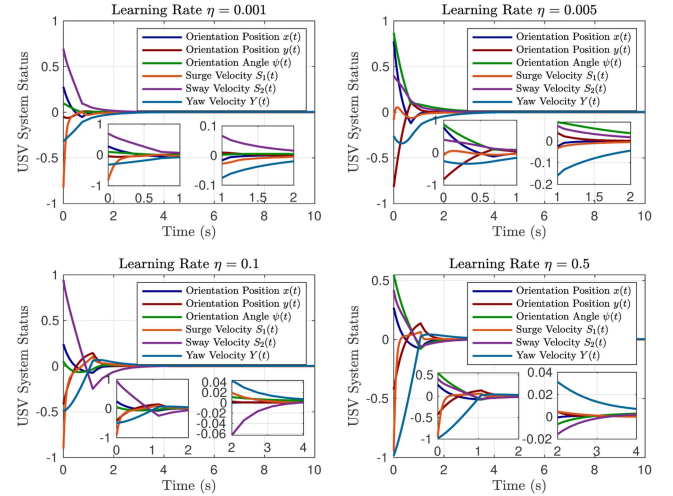


Fig. 6. USV trajectory under different learning rates.

or reinforcement learning approaches are not considered in this comparison, as the objective here is to assess communication load under identical control structures rather than policy learning performance. As the sampling period increases, all schemes exhibit a decreasing number of communication signals, which is consistent with the behavior of sampled-data and event-triggered control systems.

Under nominal conditions, the fixed-window ACK bundling strategy achieves communication loads comparable to, and in some cases lower than, those of the reference methods. When gradient-based adaptation is enabled, the communication load is further reduced in most scenarios. Under data poisoning attacks, communication activity increases for all schemes due to feedback corruption and disturbance-induced triggering. Despite this degradation, the proposed gradient-based ACK bundling generally requires fewer or comparable transmissions compared to its fixed-window counterpart. This observation indicates that adaptive regulation of the bundling window T_b remains effective in suppressing redundant ACKs even under adversarial interference, thereby preserving communication efficiency while sustaining closed-loop operation.

The results indicate that adaptive adjustment of the bundling window T_b effectively suppresses redundant ACKs while preserving stable closed-loop behavior. Under data poisoning attacks, the overall communication activity increases for all schemes due to corrupted feedback and disturbance-induced triggering. Despite this degradation, the gradient-based ACK bundling strategy generally generates fewer or comparable transmissions compared to the fixed-window design, showing that adaptive bundling remains effective in mitigating unnecessary communication under adversarial interference and in maintaining a favorable balance between communication load and control performance.

Fig. 6 illustrates representative closed-loop state trajectories under different parameter settings of the gradient-based adaptation. The adjustment rate influences the transient evolution of the bundling window T_b , which in turn affects short-term

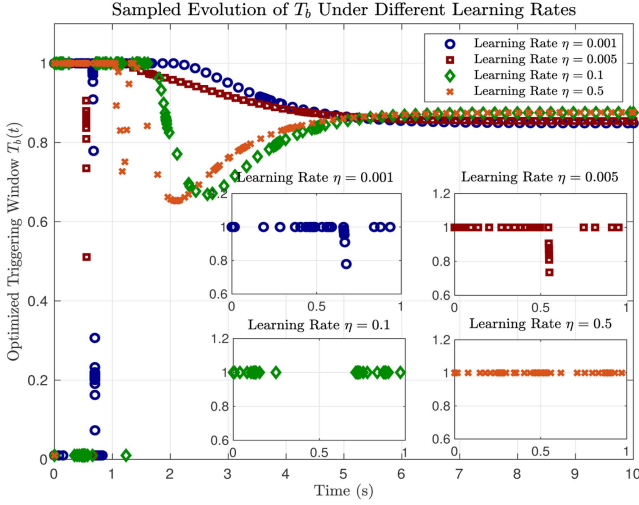
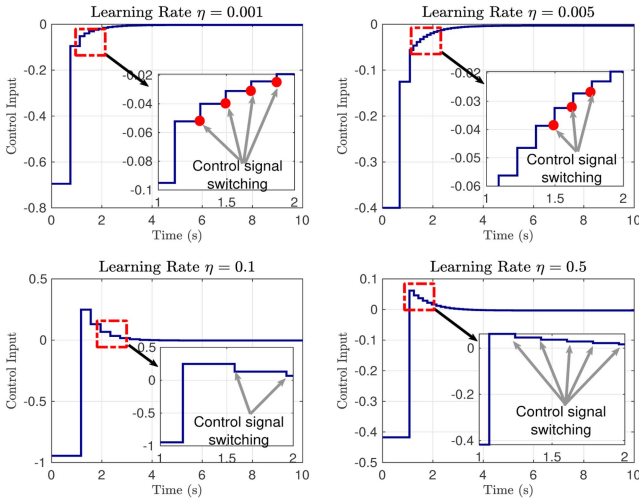
Fig. 7. Evolution of the bundling window $T_b(t)$ under different learning rates.

Fig. 8. Control input trajectories under learning rates.

communication dynamics. Moderate update rates lead to smooth T_b evolution and regular state responses, while overly aggressive or overly conservative settings introduce slower adaptation or transient oscillations. In all cases, T_b is updated online at each sampling instant from a feasible initial value. Since the bundling window directly regulates ACK frequency, these differences mainly affect transient communication behavior rather than the underlying control structure or the steady-state performance.

Fig. 7 shows the representative evolution of the ACK bundling window T_b under different parameter settings of the gradient-based adaptation. The bundling window converges smoothly under conservative update settings, whereas more aggressive settings introduce faster changes accompanied by transient fluctuations, particularly during the early adjustment stage. These variations indicate that the adaptation rate mainly affects the transient regulation of ACK timing. Fig. 8 presents the corresponding control input trajectories. Differences in the adaptation rate primarily influence short-term input smoothness through the transient behavior of T_b . Moderate update settings yield

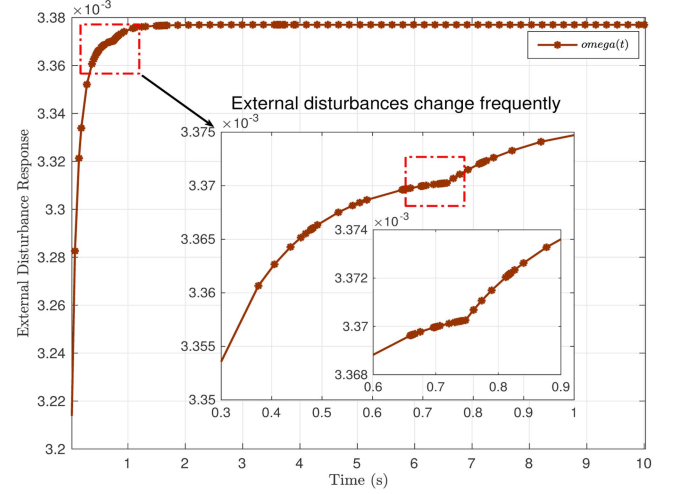
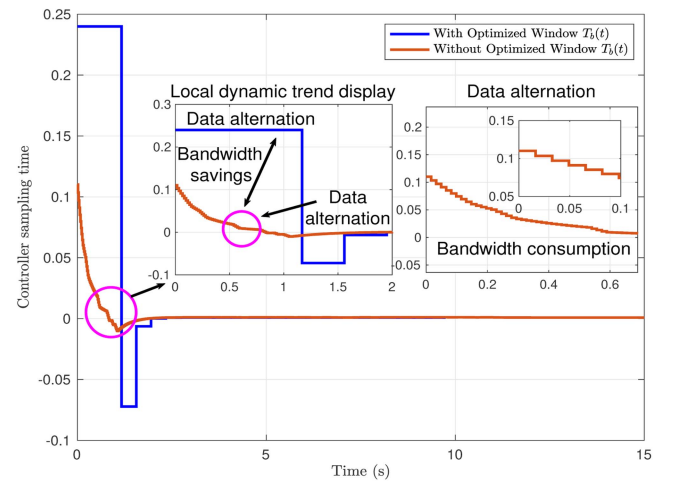


Fig. 9. Representative trajectory of external disturbances.

Fig. 10. Adaptive evolution of the bundling window $T_b(t)$ under disturbance conditions.

smoother actuation signals, while overly aggressive ones introduce noticeable oscillations during the initial phase. In all cases, the underlying control structure remains unchanged, and the observed differences are confined to transient communication-driven effects rather than the steady-state control performance.

Fig. 9 shows a representative profile of external disturbances acting on the closed-loop system, including environmental variations and adversarial perturbations. These disturbances are applied consistently across different experimental settings to evaluate system behavior under non-ideal and time-varying conditions. Fig. 10 compares the corresponding control input trajectories with and without adaptive supervision. When the bundling window T_b is adjusted online, the control inputs remain smoother with reduced oscillations, indicating improved feedback coordination and more timely actuation. In contrast, unsupervised operation leads to larger fluctuations and slower responses, highlighting the stabilizing effect of the proposed adaptive regulation under disturbance and delay.

TABLE VI
OBSERVED SYSTEM BEHAVIOR UNDER DIFFERENT COMMUNICATION DELAY THRESHOLDS

Metrics	Communication Delay Threshold d			
	1.376	1.633	2.319	> 2.319
Response Time (s)	Stable	Stable	Stable	Unstable
	19.73	20.91	32.11	Not convergent

Notably, the learning rate η is not merely an implementation parameter, but a key factor affecting the robustness, sensitivity, and adaptability of the proposed framework. A relatively large η improves the responsiveness of online adjustment, but may also induce oscillatory T_b evolution and reduce the closed-loop stability margin under corrupted feedback, revealing higher sensitivity to tuning inaccuracies. In contrast, a relatively small η leads to smoother adjustment and improved transient robustness, yet weakens the ability of the adaptive mechanism to respond to time-varying operating conditions. Therefore, the experimental results indicate that a moderate η provides a desirable balance among attack resilience, communication-aware adaptation, and closed-loop stability, thereby mitigating the instability risk caused by overly aggressive or overly conservative parameter updates.

B. System Dynamic Response Under Communication Delay

Following the learning-rate analysis, this subsection examines the impact of communication delay on closed-loop system performance. In networked control systems, excessive latency can significantly impair feedback effectiveness, potentially leading to reduced responsiveness or instability. To quantify delay tolerance, the communication delay is gradually increased under a fixed controller and ACK bundling strategy, and the resulting system behavior is evaluated.

Table VI summarizes the system response under different communication delays. Stable operation is preserved for delays up to $d = 2.319$ s, although increased latency leads to slower convergence and longer execution times. When the delay exceeds this threshold, the closed-loop system fails to converge, indicating a loss of stability. This observation confirms the existence of a critical delay bound and underscores the need for delay-aware design in networked control mechanisms such as ACK bundling.

Fig. 11 illustrates the evolution of system trajectories as communication delay increases. As latency increases, the response converges more slowly and exhibits more pronounced oscillatory behavior. While moderate delays still allow the system to track the reference trajectory, excessive delay results in pronounced instability and eventual divergence, consistent with the threshold behavior observed in Table. Fig. 12 further depicts the corresponding control input signals. For minor delays, the inputs remain smooth and well regulated, indicating effective feedback coordination. In contrast, significant delays introduce abrupt, oscillatory control actions, reflecting the growing mismatch between delayed measurements and the actual system states. These results demonstrate that the proposed framework

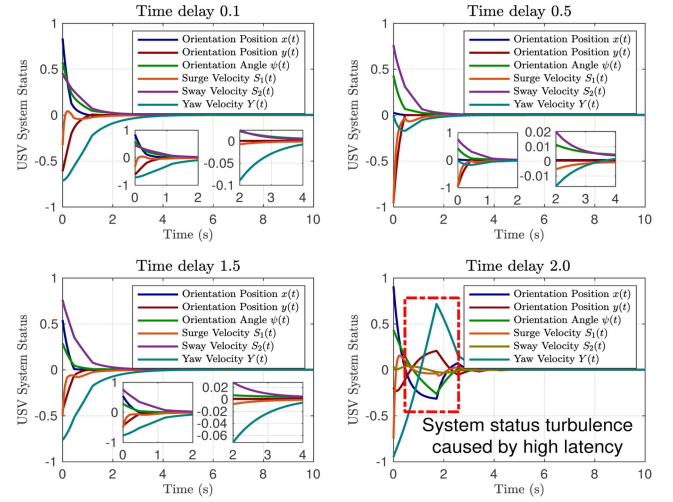


Fig. 11. USV trajectory under communication delays.

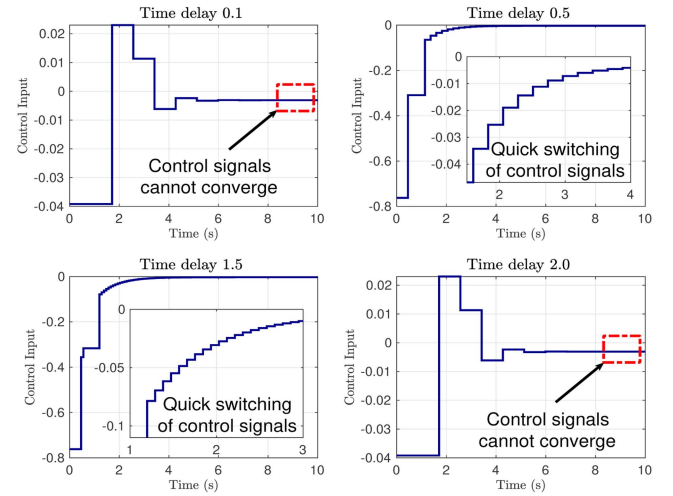


Fig. 12. Control input under different communication delays.

remains effective under AI-driven feedback poisoning and communication delays, which are the two adverse conditions most directly relevant to the present study. Broader fault scenarios, such as sensor degradation and actuator malfunctions, are beyond the current scope and will be considered in future work.

C. System Response under AI-Driven Data Poisoning Attacks

This subsection evaluates closed-loop performance under AI-driven feedback poisoning attacks. Real-world attack traces from the SWaT dataset are injected into the feedback channel, producing corrupted measurements and timing irregularities consistent with practical cyber-physical attacks. The goal is to verify whether adaptive ACK bundling can maintain tracking stability and actuation consistency under adversarial feedback manipulation.

Fig. 13 shows the USV trajectory with adaptive ACK supervision enabled. Stable tracking is preserved with only minor transient deviations under poisoned feedback. Fig. 14 shows

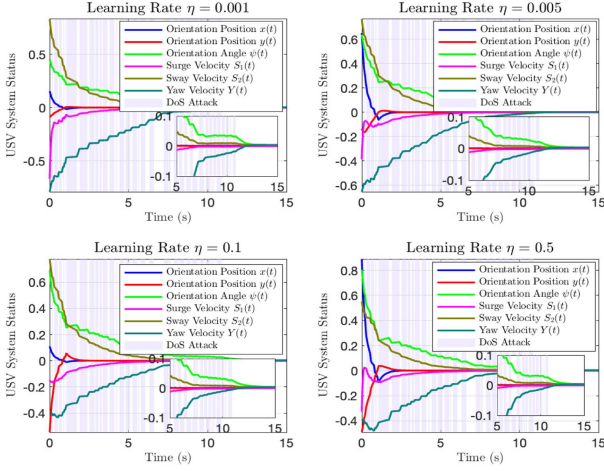


Fig. 13. USV trajectory under AI-driven data poisoning with adaptive supervision.

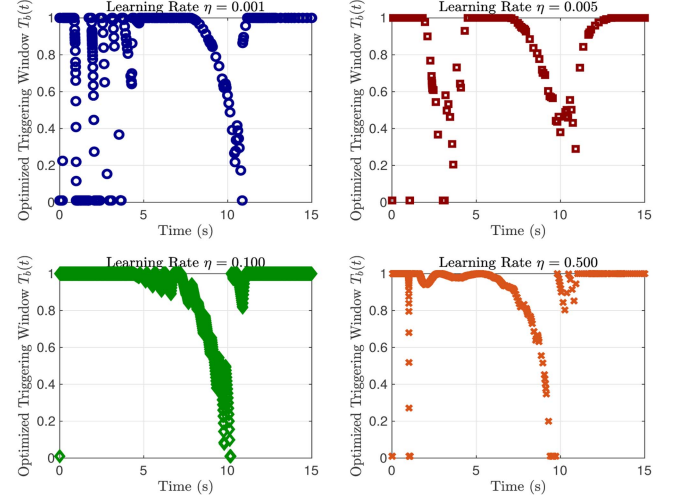


Fig. 15. $T_b(t)$ evolution under AI-driven data poisoning.

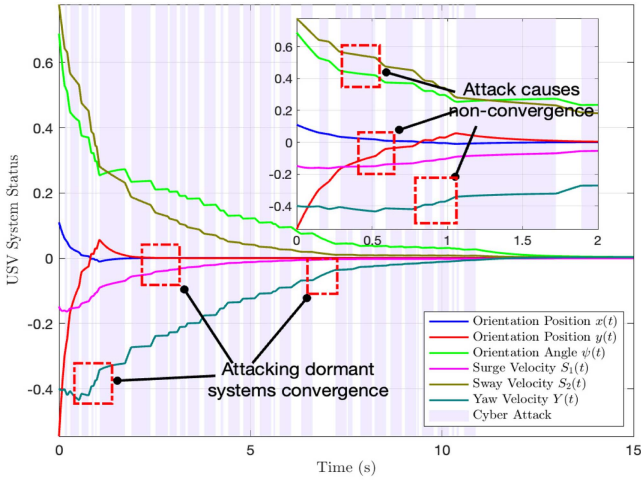


Fig. 14. USV trajectory under AI-driven data poisoning without supervision.

the unsupervised case, where fixed feedback regulation leads to amplified oscillations and visible tracking degradation. These results suggest that robustness under poisoning depends on well-regulated online adaptation of T_b . When the update becomes overly reactive to corrupted feedback, fluctuations in T_b can propagate to the actuation signal and deteriorate tracking. In contrast, restrained adaptation stabilizes feedback timing and suppresses disturbance amplification in the control loop.

Fig. 15 depicts the evolution of the ACK bundling window T_b under data poisoning attacks. When the adaptation process is properly regulated, T_b remains bounded and well behaved despite corrupted feedback, demonstrating that adversarial perturbations are absorbed at the communication layer. Conversely, insufficiently damped adaptation amplifies feedback corruption and leads to pronounced transient fluctuations, highlighting the importance of restrained adjustment under adversarial conditions. The corresponding control input responses are shown in Fig. 16. With adaptive ACK regulation, the control inputs

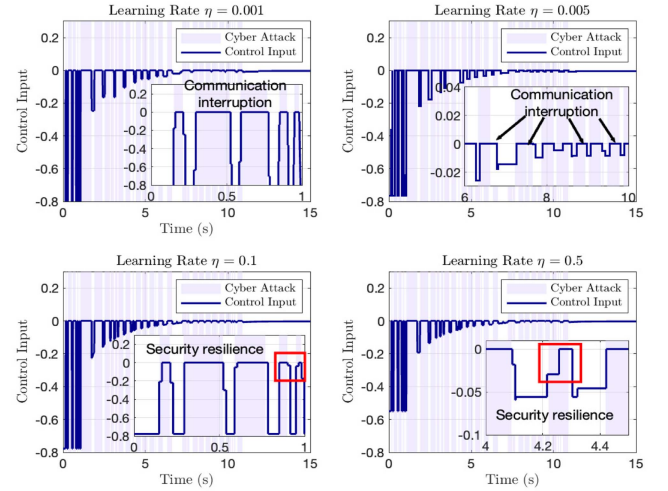


Fig. 16. Control input response under AI-driven data poisoning for different learning rates.

remain smooth and coordinated, indicating effective suppression of feedback corruption at the actuation level. In contrast, overly reactive adjustment causes poisoned feedback to propagate into abrupt and irregular control actions, thereby degrading input smoothness and increasing actuation variability. These results show that maintaining stable actuation under attack depends on moderated feedback regulation rather than aggressive parameter updates.

Fig. 17 presents the control input trajectories under AI-driven data poisoning with algorithmic supervision enabled. Compared with the unsupervised case, the control signals exhibit smaller fluctuation amplitudes and improved smoothness, reflecting more consistent actuation despite adversarial interference. This behavior confirms that adaptive ACK bundling stabilizes feedback timing and limits abrupt control variations under corrupted measurements.

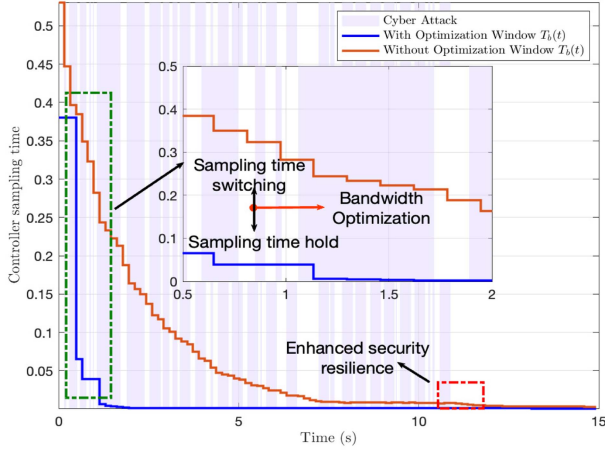


Fig. 17. Control input response under AI-driven data poisoning with adaptive supervision.

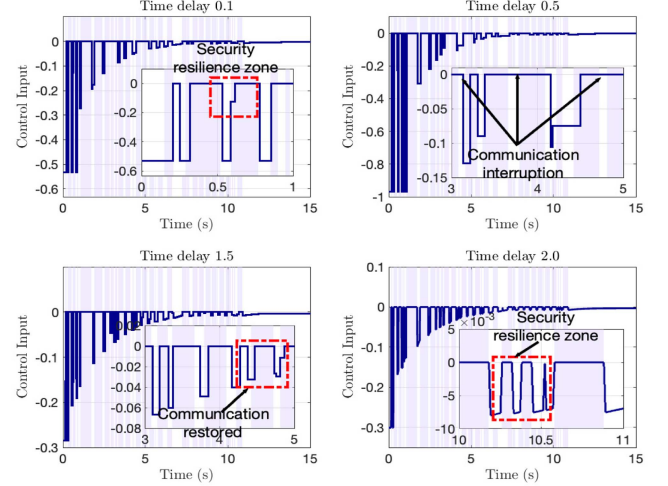


Fig. 19. Control input response under AI-driven data poisoning with different communication delays.

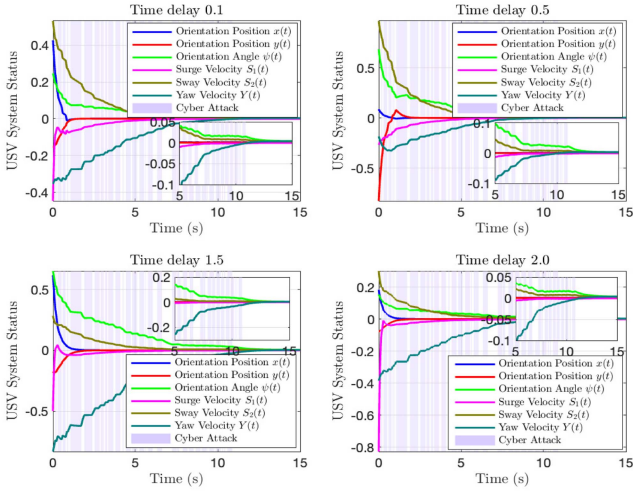


Fig. 18. USV trajectory under AI-driven data poisoning with different communication delays.

The combined effects of communication delay and data poisoning are illustrated in Figs. 18 and 19. For small delays, stable tracking and smooth control inputs are preserved despite adversarial interference. As the delay increases, oscillations become more pronounced and convergence slows. When the delay exceeds a critical threshold, trajectory divergence and loss of control coordination are observed, indicating that latency significantly amplifies the destabilizing effect of poisoned feedback.

Finally, Fig. 20 illustrates the temporal distribution of the AI-driven data poisoning attacks used in the experiments, including both attack durations and inter-attack intervals. The attacks exhibit clear temporal irregularity, ranging from brief intermittent disruptions to prolonged clustered interference, which reflects realistic adversarial behavior in cyber-physical systems. Combined with the results in Figs. 18–20, these observations provide a more comprehensive validation of the proposed framework from the perspectives of tracking robustness, adaptive ACK regulation, and control smoothness. Across all attack scenarios,

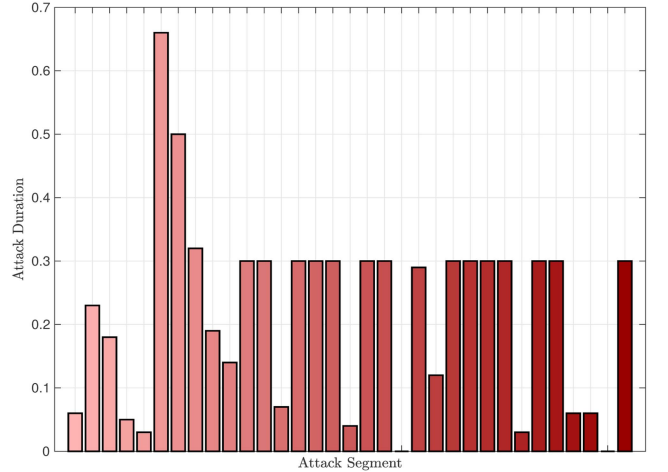


Fig. 20. Attack duration and segment interval extracted from the SWaT dataset.

the proposed mechanism consistently mitigates feedback corruption and preserves closed-loop performance under nonuniform and timing-aware adversarial conditions.

VI. CONCLUSION

This paper has investigated resilient secure control for networked industrial control systems under communication delays and AI-driven feedback poisoning attacks. A data-driven adversarial modeling approach based on real-world cyber-physical traces has been adopted to capture intelligent feedback manipulation beyond idealized threat assumptions. To address timing-aware vulnerabilities and communication overhead, an adaptive ACK bundling mechanism has been developed, in which the bundling window T_b is dynamically optimized via GD. This design enables a flexible balance between feedback synchronization and communication efficiency in adversarial network environments. In addition, a lightweight LKF has been formulated to establish closed-loop stability under delayed

and corrupted feedback with reduced analytical complexity. Simulation results on a networked USV platform demonstrate improved tracking performance, smoother control behavior, enhanced delay tolerance, and strong robustness against intelligent data poisoning attacks. Future work will focus on extending the proposed communication-layer defense to broader secure control settings, including its integration with policy-learning or RL-based control frameworks, as well as incorporating more general fault scenarios and practical implementation constraints such as computational burden, sensor noise, hardware limitations, and real-time requirements. Overall, the proposed framework provides a practical and analytically sound basis for advancing resilient secure control of networked industrial control systems in adversarial network environments.

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